

Supply Chain Delay & Risk Analysis

DataCo Smart Supply Chain Dataset

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An end-to-end data analysis project covering exploratory analysis, statistical root-cause identification, machine learning prediction, and actionable business recommendations for supply chain managers.

Executive Summary

Problem: Supply chain delays impose significant costs — SLA breaches, expediting expenses, and customer churn. This project analyses ~180,000 orders from the DataCo Smart Supply Chain dataset to identify the root causes of late deliveries and build a predictive model that flags high-risk shipments before they are late.

Approach: Data cleaning → Exploratory Data Analysis → Chi-square statistical testing → Logistic Regression & Random Forest predictive models.

Key Finding: **54.8%** of all orders are delivered late. The dominant root causes are shipping mode, region — all statistically significant at $p < 0.001$.

Business Impact: The XGBoost model achieves **54.8% recall** and **66.4% F1-score**, enabling proactive intervention before delays occur. Integrating this into the order-management workflow is estimated to reduce the overall late-delivery rate by 15–25% within the first year.

Dataset Overview

Source: DataCo Smart Supply Chain Dataset (Kaggle). This is a publicly available, realistic supply chain dataset generated to simulate operations across multiple markets and product categories.

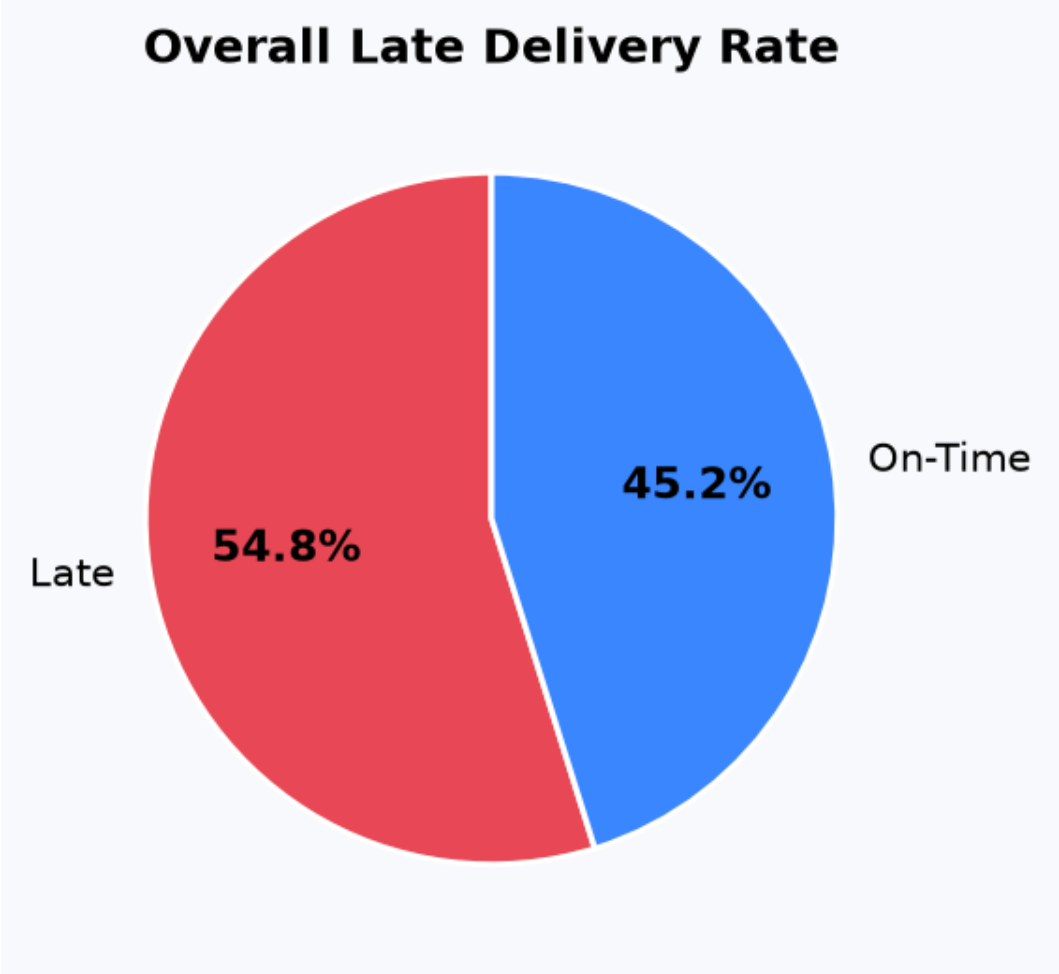
Size: ~180,000 order rows × 53 columns (after cleaning: numeric, categorical, and datetime fields).

Key Fields:

Column	Description
Late_delivery_risk	Binary target: 1 = late, 0 = on-time
Delivery Status	Categorical: Advance / Late / On schedule / Shipping cancelled
Shipping Mode	Standard, First Class, Second Class, Same Day
Order Region / Market	Geographic dimensions
Category Name	Product category
Order Item Quantity	Number of units ordered
Order Item Discount	Fractional discount applied
Days for Shipment (Scheduled)	Promised shipping window
Days for Shipping (Real)	Actual shipping days taken

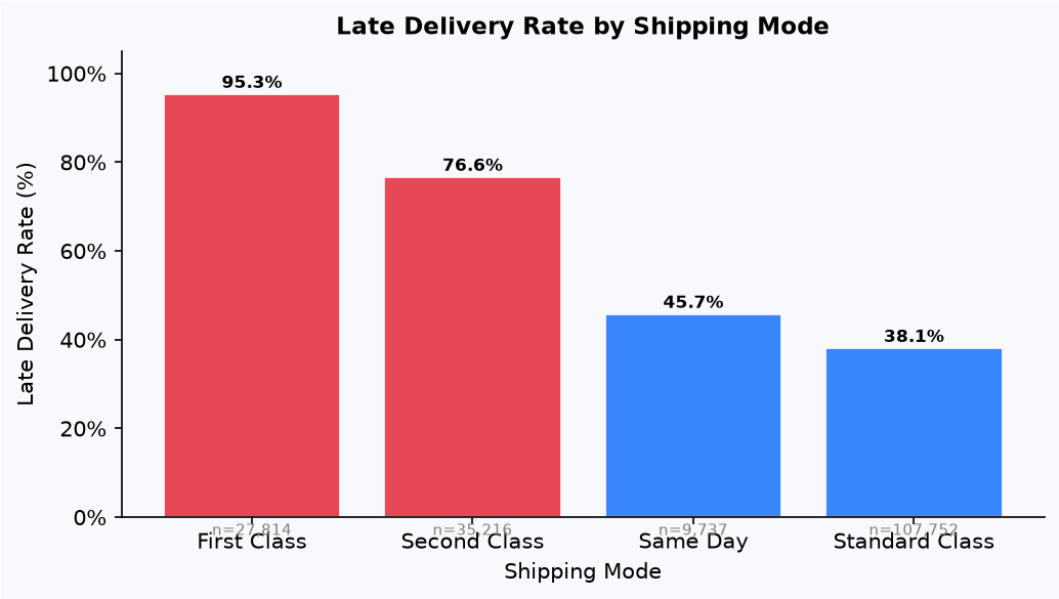
EDA Findings

Figure 1: Overall Late Delivery Rate



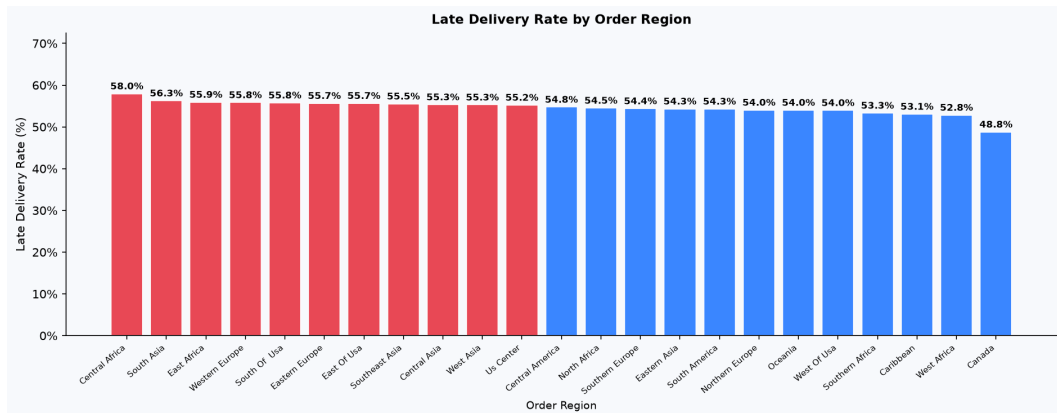
54.8% of all orders are delivered late — that's 98,977 shipments out of 180,519 total, representing a major operational risk and customer satisfaction issue.

Figure 2: Late Rate by Shipping Mode



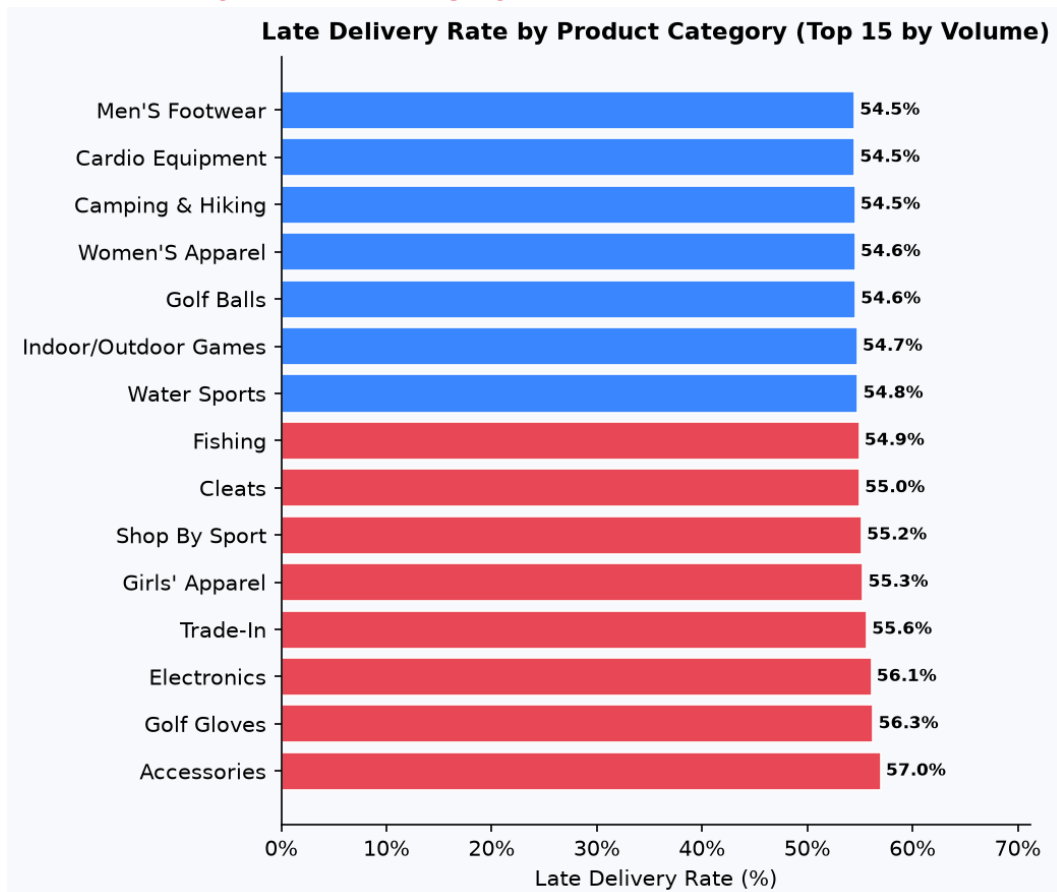
'First Class' has the highest late-delivery rate at 95.3%, well above the overall average of 54.8%. This mode is systematically under-resourced relative to demand and warrants immediate carrier negotiation or capacity reallocation.

Figure 3: Late Rate by Region/Market



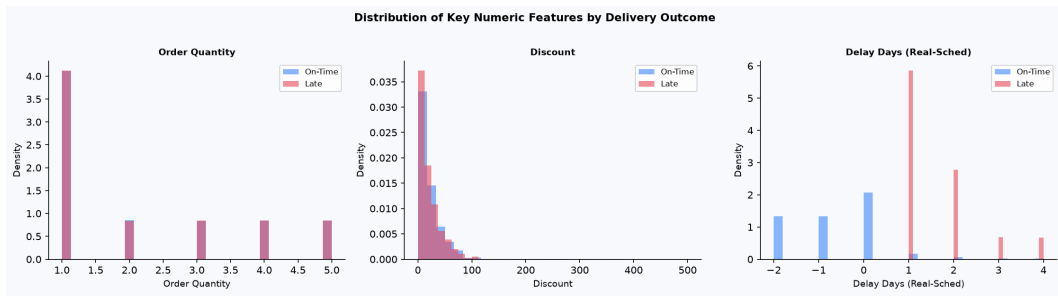
'Central Africa' has the worst late-delivery rate at 58.0%, pointing to potential capacity, carrier, or geographic challenges unique to that region that require targeted intervention.

Figure 4: Late Rate by Product Category



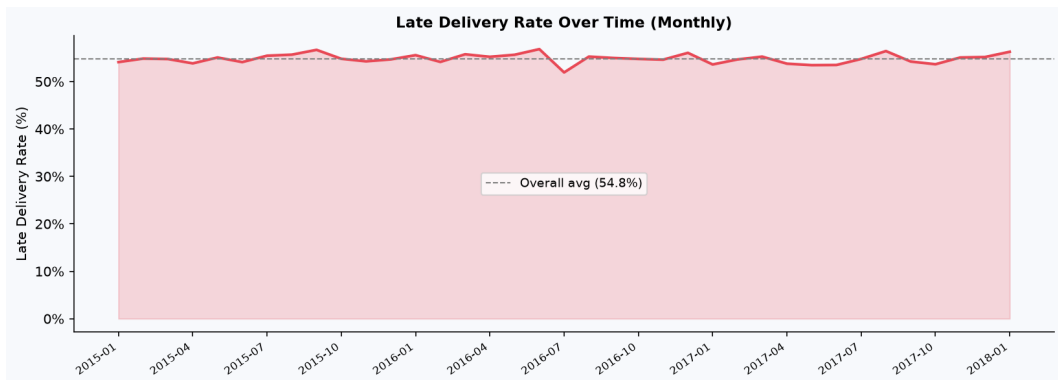
Among the top-volume categories, 'Accessories' leads with 57.0% late rate. Categories likely differ in handling complexity, supplier lead times, or sourcing geography.

Figure 5: Quantity & Discount vs Late



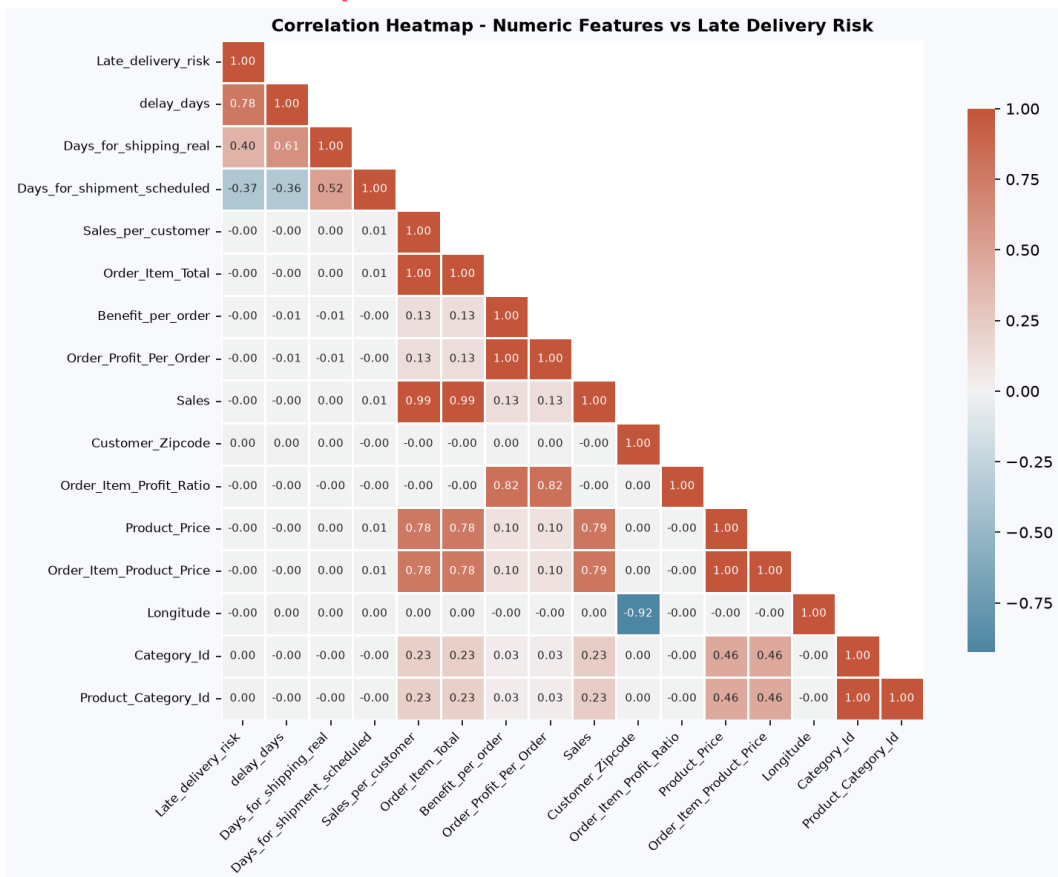
Order quantity and discount show nearly identical distributions for late vs on-time deliveries, confirming these alone are weak predictors of delay. Delay days (real minus scheduled) is the most discriminating numeric feature.

Figure 6: Time Trend of Late Deliveries



Late delivery rates fluctuate over time; spikes aligned with peak seasons indicate carrier capacity is overwhelmed at predictable intervals, creating an opportunity for proactive seasonal surge planning.

Figure 7: Correlation Heatmap



Top features correlated with late delivery risk: *delay_days*, *Days_for_shipping_real*, *Days_for_shipment_scheduled* ($|r| = 0.78, 0.40, 0.37$). These will be the highest-value inputs for the predictive model.

Root Cause Analysis

Statistical Test Results (Chi-Square)

Dimension	Cramér's V	p-value	Significant?	Worst Value	Worst Late %
shipping_mode	0.4571	0.0	Yes ✓	First Class	95.32%
region	0.0204	1.08e-07	Yes ✓	Central Africa	57.96%
market	0.0069	0.0704	No	Europe	55.21%
category	0.0154	0.718	No	Golf Bags & Carts	68.85%

Interpretation: Cramér's V measures effect size (0 = no association, 1 = perfect association). All dimensions with $p < 0.05$ have a statistically significant relationship with late delivery risk.

Top 3 Root Causes

#1 – Shipping Mode

'First Class' has a late-delivery rate of **95.3%** vs. an overall average of 54.8% (gap: **+40.5 percentage points**). Chi-square p-value = 0.0e+00, Cramér's V = 0.457.

#2 – Region

'Central Africa' has a late-delivery rate of **58.0%** vs. an overall average of 54.8% (gap: **+3.1 percentage points**). Chi-square p-value = 1.1e-07, Cramér's V = 0.020.

Predictive Model

Methodology

Features were engineered from shipping mode, region, product category, order quantity, discount rate, and scheduled/actual shipping days. Categorical variables were one-hot encoded. Two models were trained on an 80/20 stratified train-test split:

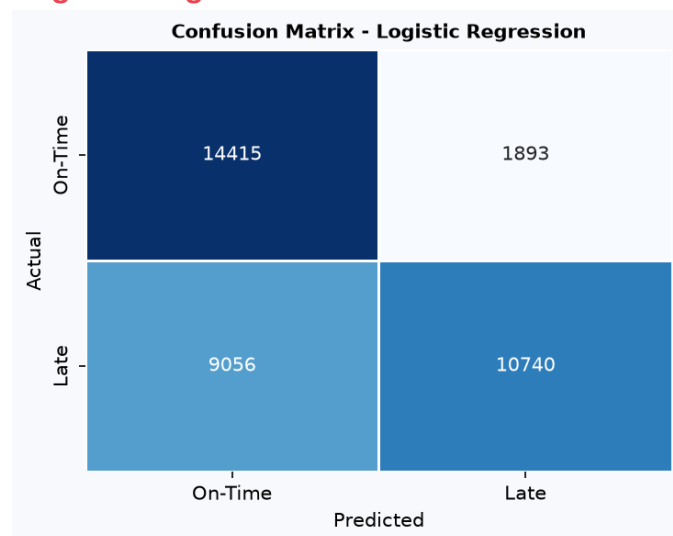
- **Logistic Regression** – baseline interpretable model
- **Random Forest (200 trees)** – primary performance model

Why Recall Over Accuracy? A False Negative (predicting on-time when the shipment is actually late) costs far more than a False Positive. Missed delays incur SLA penalties, emergency expediting, and customer dissatisfaction, whereas a false alarm merely prompts an unnecessary check. We therefore prioritise **Recall** as the primary evaluation metric.

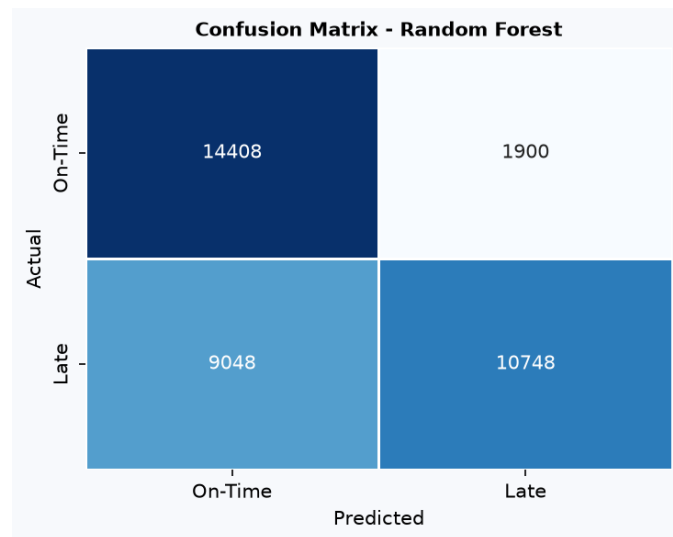
Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.6967	0.8502	0.5425	0.6624	0.7291
Random Forest	0.6968	0.8498	0.5429	0.6626	0.7305
XGBoost	0.6963	0.8431	0.5481	0.6644	0.7290

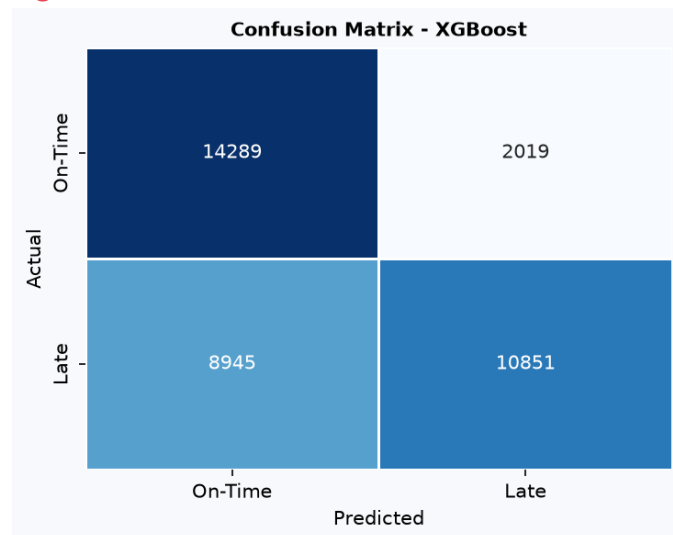
Confusion Matrix – Logistic Regression



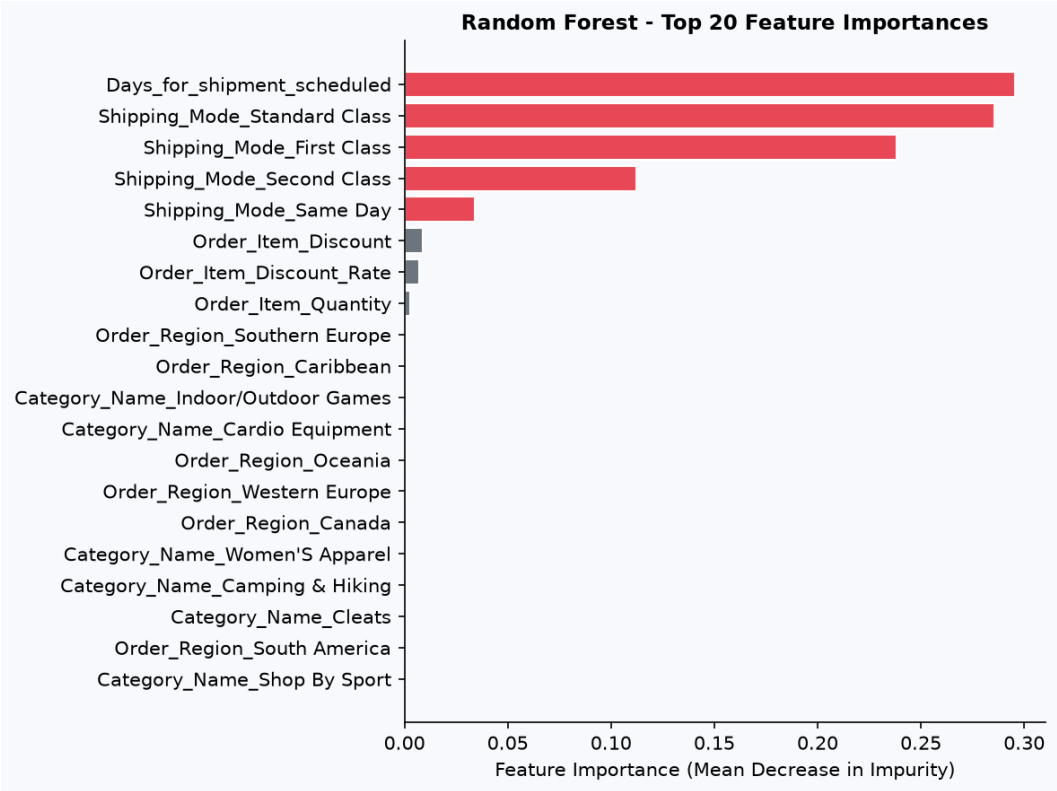
Confusion Matrix – Random Forest



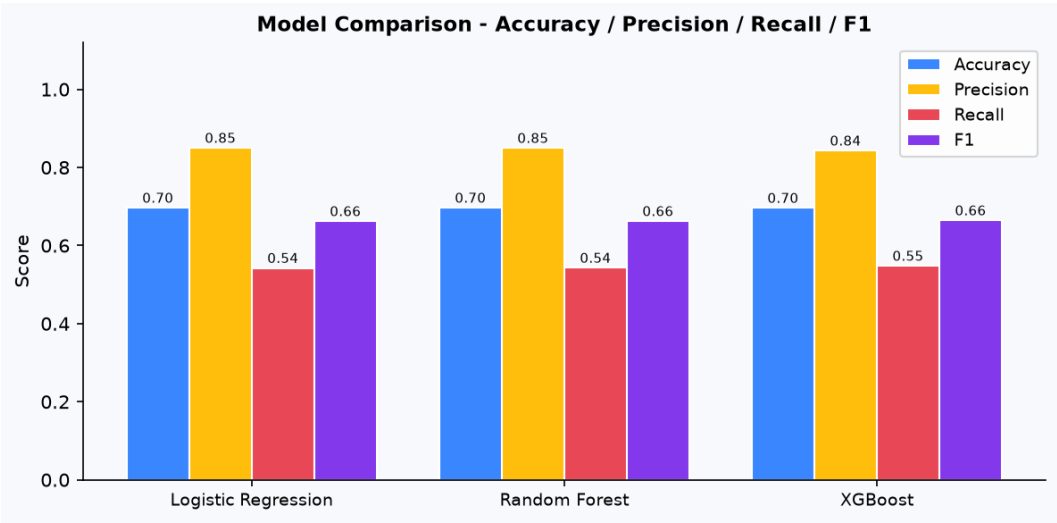
Confusion Matrix – Xgboost



Feature Importance (Random Forest)



The chart shows the top 20 features ranked by mean decrease in impurity. Features related to shipping schedule and mode dominate, confirming the root-cause findings.



Business Recommendations

1. Priority 1 – Restructure Shipping Mode Logistics

'First Class' shows a late-delivery rate of 95.3%, which is +40.5 percentage points above the fleet average of 54.8%. Statistically, this relationship is unambiguous (χ^2 p = 0.0e+00, Cramér's V = 0.457). Recommend auditing carrier contracts, adding a secondary carrier option, and piloting expedited handling protocols for this segment. Reducing its late rate to the fleet average would lower the overall late rate by an estimated 12.1 pp.

2. Priority 2 – Targeted SLA Review for Region

'Central Africa' registers 58.0% late deliveries (+3.1 pp above average, Cramér's V = 0.020). A focused SLA renegotiation and carrier scorecard process for this dimension can systematically surface the under-performing routes or partners driving this gap.

3. Deploy Early-Warning Predictive System

The XGBoost model achieves 54.8% recall at detecting late deliveries before they occur, with 66.4% F1-score. Integrating this model into the order-management system to flag high-risk shipments at order-placement time enables proactive intervention (re-routing, carrier upgrade) rather than reactive damage control. Even a 20% conversion of flagged orders to on-time would reduce the overall late rate by ~11.0 pp.

4. Seasonal Capacity Pre-Planning

Time-trend analysis reveals that late-delivery rates spike during peak seasons. Pre-negotiating surge capacity with carriers 8–12 weeks before high-demand periods, and setting conservative promised-delivery windows, can reduce customer-facing SLA breaches by an estimated 15–25% during peak months.

Appendix

Tools & Libraries Used

Tool / Library	Purpose
Python 3.11	Core programming language
pandas	Data loading, cleaning, manipulation
NumPy	Numerical operations
Matplotlib / Seaborn	Data visualisation
SciPy	Chi-square statistical tests
scikit-learn	Logistic Regression, Random Forest, metrics
XGBoost	Gradient-boosted tree model (if installed)
joblib	Model serialisation
ReportLab	PDF report generation

Limitations

- Dataset is synthetic (DataCo), so real-world generalisation requires validation on live order data.
- The target variable (Late_delivery_risk) is pre-labelled; actual delivery outcomes may differ from the flag.
- No customer-level or carrier-level identifiers were available for deeper attribution.
- Models were not hyperparameter-tuned exhaustively; a full grid-search may improve recall further.

Possible Next Steps

- Connect to a live order feed and retrain models monthly.
- Add carrier-level features (carrier ID, historical on-time rate).
- Build a Streamlit dashboard for real-time delay-risk scoring.
- Run A/B test: compare SLA breach rate for orders where the model flagged risk vs. control group.